Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 08)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

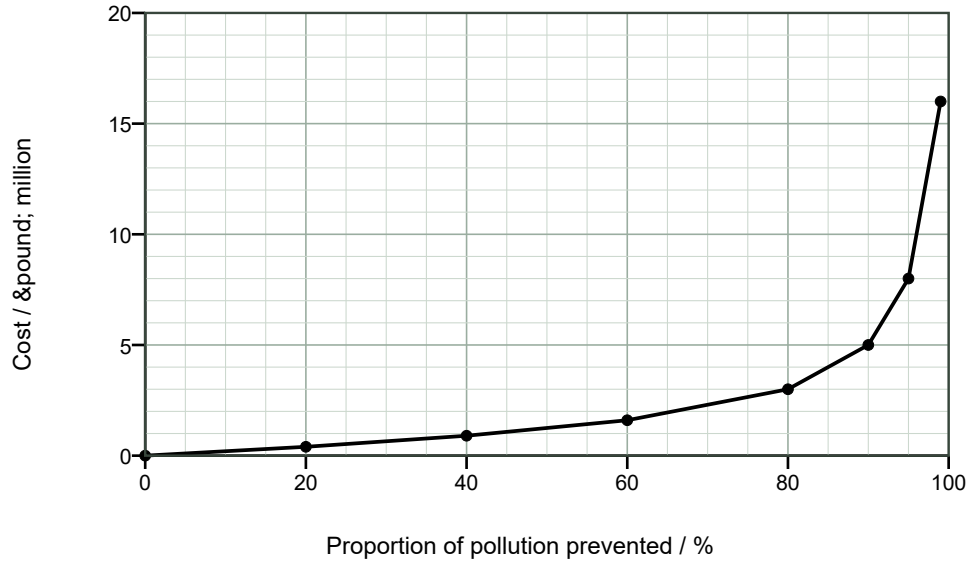
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



0 1 . 1 Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer £ million

0 1 . 2 Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

.....

.....

.....

.....

.....

.....

Do not write outside the box

0 1 . 3 Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

0 1 . 4 Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

15

0 2 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

0 2 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

.....

.....

.....

.....

0 2 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

.....

.....

.....

.....

5

0 3 . 1 Complete Table 1 to explain how each method reduces soil erosion.

[5 marks]

Table 1

Method	How it reduces erosion	
Terracing		
Tied ridging		
Windbreaks / hedgerows		
Mulching		
Strip cropping		

.....

Do not write outside the box

Do not write outside the box

0 3 . 2 Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops.

[2 marks]

0 3 . 3 Describe **three** ways livestock can be managed to reduce soil erosion.

[3 marks]

1

2

3

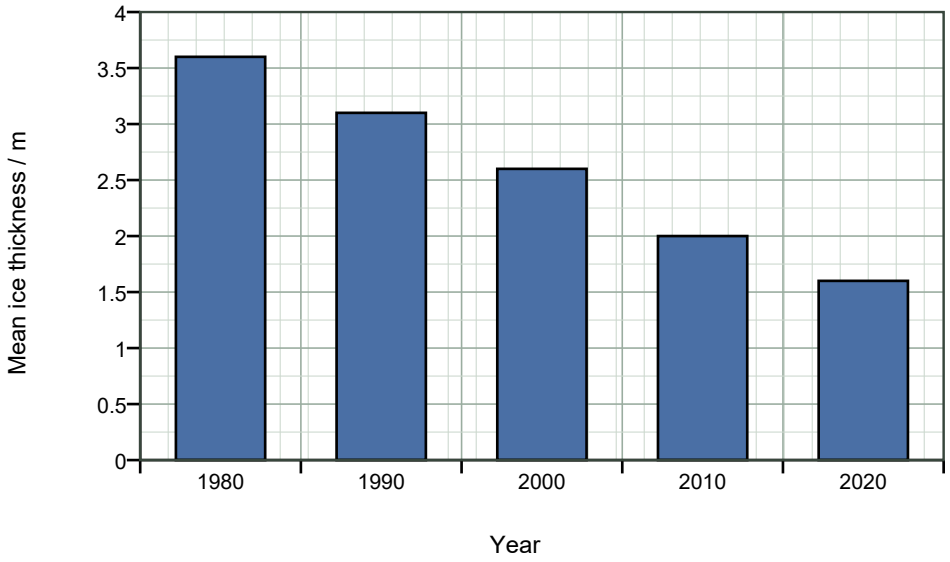
10

0 4 Table 1 shows the area of Arctic sea ice in September (the annual minimum) in selected years. Figure 1 shows the mean thickness of Arctic sea ice in the same years.

Table 1

Year	1980	1990	2000	2010	2020
September sea ice area / million km^2	7.7	6.2	6.3	4.9	3.9

Figure 1



0 4 . 1 Calculate the percentage decrease in September sea ice area between 1980 and 2020.

[2 marks]

Answer ____ %

0 4 . 2 Explain why the loss of sea ice does **not** directly raise sea level, but the loss of land ice does.

[2 marks]

.....

.....

.....

Do not write outside the box

Do not write outside the box

.....

0 4 . 3 Explain how the changes shown in Table 1 and Figure 1 could lead to further warming.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4 . 4 Explain **two** ways in which rising temperatures cause sea level to rise.

[2 marks]

.....

.....

.....

.....

0 4 . 5 Explain why the break-up of an ice shelf can increase the rate at which land ice flows into the sea.

[1 mark]

.....

.....

10

0 5 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 5 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

Do not write outside the box

.....

.....

.....

.....

0 5 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

1 .....

.....

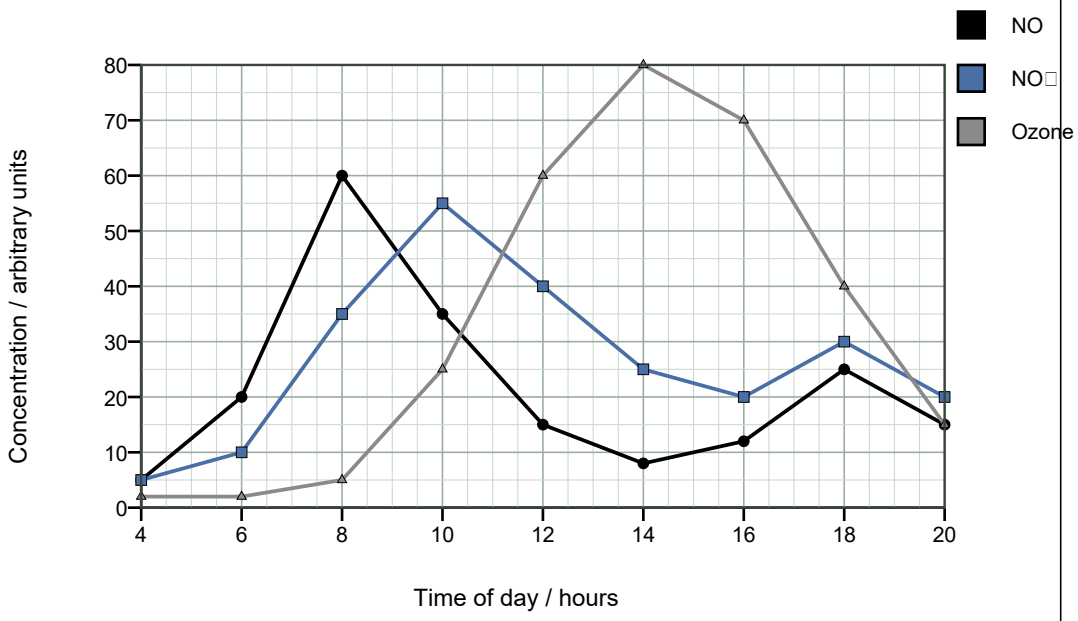
2 .....

.....

5

0 6 Figure 1 shows the concentrations of three pollutants in the air of a large city in a valley during one sunny day.

Figure 1



0 6 . 1 Use Figure 1 to describe how the ozone concentration changes during the day and explain the pattern.

[4 marks]

.....

.....

.....

Do not write outside the box

0 6 . 2 Explain why the early-morning peak in NO is followed by a peak in NO_2.

[2 marks]

0 6 . 3 Name **one** other toxic secondary pollutant produced in photochemical smogs and give one of its effects on humans.

[2 marks]

0 6 . 4 Explain why cities such as Los Angeles and Mexico City are particularly prone to photochemical smogs.

[2 marks]

10

0 7 . 1 Explain why mineral extraction causes more land-use conflict than most other human activities.

[2 marks]

0 7 . 2 Describe how each of the following environmental impacts of an operating mine can be reduced.

[5 marks]

Dust

Noise

Turbid drainage water

Acidic leachate from spoil

Loss of amenity

0 7 . 3 Explain how spoil heaps can be made stable and describe what happened at Aberfan in 1966.

[3 marks]

Do not write outside the box

10

0 8 . 1 Evaluate the risks and benefits of the uses of ionising radiation, and explain the principles used to control exposure of workers and the public.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

0 8 . 2 Give **one** reason why the risks and benefits of nuclear electricity may not fall on the same people.

[1 mark]

10

0 9 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 9 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

.....

.....

.....

.....

0 9 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

.....

.....

.....

.....

5

1 0 A student investigated whether the number of earthworms per m^2 differs between an organic field and a conventionally farmed field. Table 1 shows the results.

Table 1

Field	Number of samples	Mean earthworms per m^2	Standard deviation
Organic	10	48	14
Conventional	10	31	12

1 0 . 1 Name the statistical test the student should use and explain why it is appropriate.

[2 marks]

.....

.....

Do not write outside the box

Do not write outside the box

1 0 . 2 The test gave p = 0.01. State the conclusion the student should draw.

[2 marks]

1 0 . 3 Explain why the student cannot conclude that organic farming **caused** the higher earthworm numbers.

[2 marks]

1 0 . 4 Evaluate the decisions that must be made when planning a fieldwork investigation so that valid conclusions can be drawn.

[9 marks]

Do not write outside the box

Extra space

15

Do not write outside the box

Write an essay on one of the following topics.

1 1 . 1 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

1 1 . 2 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS